

Supporting Information

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SI Derivation of the Maximum Efficiency Without Thomson Heat

The conversion efficiency of a homogeneous TE leg at a given temperature difference is expressed as a function of the ratio of an external electric resistance R_L to internal electric resistance R , $m = R_L/R$, as

$$\eta = \frac{P_{out}}{Q_{in}} = \frac{\frac{V_{oc}^2}{R} \frac{m}{(1+m)^2}}{\frac{A}{L} \int_{T_c}^{T_h} \kappa(T) dT + IT_h S(T_h) - \frac{1}{2} I^2 R}, \quad [S1]$$

where V_{oc} , κ , and $S(T_h)$ are the open-circuit voltage, thermal conductivity, and Seebeck coefficient at hot-side temperature, respectively. A and L are a cross-section area and length of a TE leg, respectively. R is expressed as

$$R = \frac{1}{\Delta T} \frac{L}{A} \int_{T_c}^{T_h} \rho(T) dT. \quad [S2]$$

By applying a load current $I = V_{oc}/(R(1+m))$ and Eq. S2 into Eq. S1, the efficiency becomes

$$\eta = \frac{\frac{m}{1+m}}{\frac{(1+m) \int_{T_c}^{T_h} \kappa(T) dT \int_{T_c}^{T_h} \rho(T) dT}{V_{oc}^2 \Delta T} + \frac{S(T_h) T_h}{V_{oc}} - \frac{1}{2(1+m)}}. \quad [S3]$$

The open-circuit voltage V_{oc} is rewritten in terms of temperature-dependent Seebeck coefficient as

$$V_{oc} = \int_{T_c}^{T_h} S(T) dT. \quad [S4]$$

Substituting Eq. S4 into Eq. S3 yields

$$\eta = \frac{\frac{m}{1+m}}{\frac{1+m}{Z_{eng} \Delta T} + \frac{S(T_h) T_h}{\int_{T_c}^{T_h} S(T) dT} - \frac{1}{2(1+m)}}. \quad [S5]$$

Here, we define an engineering dimensionless figure of merit $(ZT)_{eng}$ as

$$(ZT)_{eng} = Z_{eng} \Delta T = \frac{\left(\int_{T_c}^{T_h} S(T) dT \right)^2}{\int_{T_c}^{T_h} \rho(T) dT \int_{T_c}^{T_h} \kappa(T) dT} \Delta T = \frac{(PF)_{eng}}{\int_{T_c}^{T_h} \kappa(T) dT} \Delta T, \quad [S6]$$

where Z_{eng} is the engineering figure of merit in K^{-1} , and $(PF)_{eng}$ is the engineering power factor associated with ΔT in $W m^{-1} K^{-1}$. The maximum efficiency can be obtained at the optimized ratio m_{opt} satisfying $d\eta/dm = 0$ at a given boundary temperature, T_h and T_c . After taking the derivative of Eq. S5 with respect to m and setting the result equal to zero, m_{opt} is obtained as

$$m_{opt} = \sqrt{1 + (ZT)_{eng} \left(\frac{\hat{\alpha}}{\eta_c} - \frac{1}{2} \right)}, \quad [S7]$$

$$\hat{\alpha} = \frac{S(T_h) \Delta T}{\int_{T_c}^{T_h} S(T) dT}. \quad [S8]$$

$\hat{\alpha}$ is defined as a nondimensional intensity factor of the Thomson effect. By substituting Eq. S7 into Eq. S5, the new formula for the maximum efficiency taking the temperature dependence of TE properties associated with ΔT into account is finalized as

$$\eta_{max} = \eta_c \frac{\sqrt{1 + (ZT)_{eng} (\hat{\alpha}/\eta_c - 1/2)} - 1}{\hat{\alpha} \left(\sqrt{1 + (ZT)_{eng} (\hat{\alpha}/\eta_c - 1/2)} + 1 \right) - \eta_c}, \quad [S9]$$

where η_c is the Carnot efficiency.

In addition, the efficiency at the maximum output power mode, i.e., $m = 1$ based on the constant property model, is given as (7)

$$\eta_{P_{max}} = \eta_c \frac{1}{\frac{4}{ZT_h} + 2 - \frac{1}{2} \eta_c}. \quad [S10]$$

Now efficiency at the maximum output power mode is redefined. By taking $dP_{out}/dm = d/dm((V_{oc}^2/R)m/(1+m)^2) = 0$, the condition $m = 1$ for maximum output power is obtained. Substituting $m = 1$ into Eq. S5 yields the efficiency when a material generates the maximum output power,

$$\eta_{P_{max}} = \eta_c \frac{1}{\frac{4\eta_c}{(ZT)_{eng}} + 2\hat{\alpha} - \frac{1}{2} \eta_c}. \quad [S11]$$

It is clearly noted that Eq. S7 can be reduced to $\sqrt{1 + Z \cdot T_{avg}}$ when $Z_{eng} = Z$ and $\hat{\alpha} = 1$; hence, the maximum efficiency formula based on $(ZT)_{eng}$ in Eq. S9 is converted to Eq. 1 (in the main text) derived from the constant property model. In the same manner, the conventional formula by Eq. S10 at the maximum output power mode assuming $m = 1$ is also obtained from Eq. S11. Therefore, the newly derived formulas shown by Eqs. S9 and S11 for the conversion efficiency are analytically generic and applicable to the models by constant as well as cumulative temperature-dependent properties of any TE materials.

SI Derivation of the Maximum Efficiency Including Thomson Heat

The governing equation for energy balance over one-dimensional differential element of TE material is (7)

$$\frac{d}{dx} \left(\kappa(T) \frac{dT}{dx} \right) + J^2 \rho(T) - J \tau(T) \frac{dT}{dx} = 0, \quad [S12]$$

where x and J are the distance from the heat source and current density, respectively, and $\tau(T)$ is temperature-dependent Thomson coefficient defined as $\tau(T) = T dS(T)/dT$. Integrating Eq. S12 with respect to x becomes

$$\kappa(T) \frac{dT}{dx} + J^2 \int_0^x \rho(T) dx - J \int_0^x \tau(T) \frac{dT}{dx} dx + C_1 = 0. \quad [S13]$$

The second time integration with respect to x yields

$$\int_{T_h}^T \kappa(T) dT + J^2 \int_0^x \int_0^x \rho(T) dx dx - J \int_0^x \int_{T_h}^T \tau(T) dT dx + C_1 x + C_2 = 0. \quad [\text{S14}]$$

By applying the boundary conditions of $T = T_h$ at $x = 0$ and $T = T_c$ at $x = L$, where L is a length in the x direction of the TE leg, the integration constants C_1 and C_2 are solved as

$$C_1 = \frac{1}{L} \int_{T_c}^{T_h} \kappa(T) dT - \frac{J^2}{L} \int_0^L \int_0^x \rho(T) dx dx + \frac{J}{L} \int_0^L \int_{T_h}^T \tau(T) dT dx, \quad [\text{S15}]$$

$$C_2 = 0. \quad [\text{S16}]$$

From Eqs. S13 and S15, the conduction heat at $T = T_h$ becomes

$$\begin{aligned} -A\kappa(T) \frac{dT}{dx} \Big|_{T_h} &= \frac{A}{L} \int_{T_c}^{T_h} \kappa(T) dT - \frac{I^2}{AL} \int_0^L \int_0^x \rho(T) dx dx \\ &+ \frac{I}{L} \int_0^L \int_{T_h}^T \tau(T) dT dx, \end{aligned} \quad [\text{S17}]$$

where A and I are the cross-sectional area and electric current, respectively. The constitutive relation for heat Q and electric current I is given as

$$Q = -A\kappa \frac{dT}{dx} + ITS. \quad [\text{S18}]$$

By substituting Eq. S17 into Eq. S18 and applying the boundary conditions, Q_h at the hot side is obtained as

$$Q_h = \frac{A}{L} \int_{T_c}^{T_h} \kappa(T) dT + IT_h S(T_h) - W_J I^2 R - W_T I \int_{T_c}^{T_h} \tau(T) dT. \quad [\text{S19}]$$

The terms on the right-hand side represent conduction heat, Peltier heat, Joule heat, and Thomson heat, respectively, where $S(T_h)$ is the Seebeck coefficient at T_h , and R is the electrical resistance of a TE leg,

$$R = \frac{1}{\Delta T} \frac{L}{A} \int_{T_c}^{T_h} \rho(T) dT. \quad [\text{S20}]$$

W_J and W_T are dimensionless weight factors of Joule and Thomson heat, respectively, which are defined as

$$W_J = \frac{\int_0^L \int_0^x \rho(T) dx dx}{\frac{L^2}{\Delta T} \int_{T_c}^{T_h} \rho(T) dT}, \quad [\text{S21}]$$

$$W_T = \frac{\int_0^L \int_{T_c}^{T_h} \tau(T) dT dx}{L \int_{T_c}^{T_h} \tau(T) dT}. \quad [\text{S22}]$$

All integral terms are key parameters to account for cumulative temperature dependence of S , ρ , and κ according to a temperature gradient. Muto et al. also showed similar expressions (31). The details regarding W_J and W_T will be discussed later. The conversion efficiency of TE materials accounting for the Thomson effect can be expressed as the ratio of output power (see the next section) to input heat rate,

$$\eta = \frac{P_{out}}{Q_h} = \frac{\frac{V_{oc}^2}{R} \frac{m}{(1+m)^2}}{\frac{A}{L} \int_{T_c}^{T_h} \kappa(T) dT + IT_h S(T_h) - W_J I^2 R - W_T I \int_{T_c}^{T_h} \tau(T) dT}, \quad [\text{S23}]$$

where m is the ratio of external to internal electrical resistance.

Substituting Eq. S20 into Eq. S23 and applying $I = V_{oc} R^{-1} (1+m)^{-1}$ to Eq. S23 yields

$$\eta = \frac{\frac{m}{1+m}}{\frac{1+m}{Z_{eng} \Delta T} + \frac{S(T_h) T_h}{\int_{T_c}^{T_h} S(T) dT} - \frac{W_J}{(1+m)} - \frac{W_T \int_{T_c}^{T_h} \tau(T) dT}{\int_{T_c}^{T_h} S(T) dT}}, \quad [\text{S24}]$$

where Z_{eng} is the engineering figure of merit, and $(ZT)_{eng}$ is the engineering dimensionless figure of merit, defined as

$$Z_{eng} = \frac{\left(\int_{T_c}^{T_h} S(T) dT \right)^2}{\int_{T_c}^{T_h} \rho(T) dT \int_{T_c}^{T_h} \kappa(T) dT} = \frac{(PF)_{eng}}{\int_{T_c}^{T_h} \kappa(T) dT}, \quad [\text{S25}]$$

$$(ZT)_{eng} = Z_{eng} \Delta T. \quad [\text{S26}]$$

By optimizing the ratio m satisfying $d\eta/dm = 0$ regarding Eq. S24, the maximum efficiency and its corresponding m_{opt} are finalized as

$$\eta_{max} = \eta_c \frac{\sqrt{1 + (ZT)_{eng} \alpha_1 \eta_c^{-1}} - 1}{\alpha_0 \sqrt{1 + (ZT)_{eng} \alpha_1 \eta_c^{-1} + \alpha_2}}, \quad [\text{S27}]$$

$$m'_{opt} = \sqrt{1 + (ZT)_{eng} \alpha_1 \eta_c^{-1}}, \quad [\text{S28}]$$

where η_c is the Carnot efficiency, and α_i is defined as

$$\alpha_i = \frac{S(T_h) \Delta T}{\int_{T_c}^{T_h} S(T) dT} - \frac{\int_{T_c}^{T_h} \tau(T) dT}{\int_{T_c}^{T_h} S(T) dT} W_T \eta_c - i W_J \eta_c. \quad (i = 0, 1, \text{ and } 2). \quad [\text{S29}]$$

Calculating correct W_J and W_T is essential to obtain an exact prediction of the maximum efficiency by Eq. S27. Because the cumulative temperature dependence of ρ and τ is considered under large temperature gradient in this model, the lumped constant 1/2 by the constant property model (7), i.e., one-half of Joule and Thomson heat to the hot side, is not exact anymore, so practical Joule and Thomson heat influence on the hot side should be evaluated. As shown in Eqs. S21 and S22, a temperature gradient dT/dx is required to calculate exact weight factors. However, because it is a solution of the nonlinear differential equation in Eq. S12 in the case that S , ρ , and κ are temperature dependent, the exact W_J and W_T cannot be directly solved. When ignoring radiation/convection heat loss and assuming the temperature gradient dominantly caused by conduc-

tion heat, one can express the temperature gradient by applying $Q_{\text{conduction}} = (A/L) \int_{T_c}^{T_h} \kappa(T) dT$ from the first term in Eq. S19 and $\kappa_{\text{avg}} = (1/\Delta T) \int_{T_c}^{T_h} \kappa(T) dT$ into the definition of conduction heat,

$$\frac{dT}{dx} \approx -\frac{Q_{\text{conduction}}}{A\kappa_{\text{avg}}} = -\frac{\frac{A}{L} \int_{T_c}^{T_h} \kappa(T) dT}{\frac{A}{\Delta T} \int_{T_c}^{T_h} \kappa(T) dT} = -\frac{\Delta T}{L}. \quad [\text{S30}]$$

Based on the relation $dx \sim -(L/\Delta T)dT$, the weight factors can be rewritten with respect to T as

$$W_J = \frac{\int_{T_c}^{T_h} \int_{T_c}^{T_h} \rho(T) dT dT}{\Delta T \int_{T_c}^{T_h} \rho(T) dT}, \quad [\text{S31}]$$

$$W_T = \frac{\int_{T_c}^{T_h} \int_{T_c}^{T_h} \tau(T) dT dT}{\Delta T \int_{T_c}^{T_h} \tau(T) dT}. \quad [\text{S32}]$$

If $\rho(T)$ and $\tau(T)$ are assumed to be temperature independent, the weight factors of Eqs. S31 and S32 become 1/2, and then Eq. S19 is equal to the form based on the constant property model (7). Thus, the W_J and W_T are good general approximations for the corrected contribution of Joule and Thomson heating on the hot side of a TE material.

SI Output Power Based on the Cumulative Temperature Dependence Model

In this section, it is proved that the output power as shown in Eqs. S1 and S23 takes the Thomson effect into account. The heat flux at hot side Q_h considering the Thomson effect is, by Eq. S19,

$$Q_h = IT_h S(T_h) + \frac{A}{L} \int_{T_c}^{T_h} \kappa(T) dT - W_J I^2 R - W_T I \int_{T_c}^{T_h} \tau(T) dT, \quad [\text{S33}]$$

and the heat flux at cold side Q_c accounting for the Thomson effect is also expressed as

$$Q_c = IT_c S(T_c) + \frac{A}{L} \int_{T_c}^{T_h} \kappa(T) dT - W_J I^2 R - W_T I \int_{T_c}^{T_h} \tau(T) dT - I \int_0^x \tau(T) \frac{dT}{dx} dx + I^2 R. \quad [\text{S34}]$$

By considering an adiabatic condition and no electrically parasitic loss, the output power generation is obtained based on energy conservation,

$$P_{\text{out}} = Q_h - Q_c = I(T_h S(T_h) - T_c S(T_c)) + I \int_0^L \tau(T) \frac{dT}{dx} dx - I^2 R. \quad [\text{S35}]$$

The second term on the right-hand side becomes

$$\begin{aligned} I \int_0^L \tau(T) \frac{dT}{dx} dx &= I \int_0^L T \frac{dS}{dT} \frac{dT}{dx} dx \\ &= I \int_{T_h}^{T_c} T \frac{dS}{dT} dT \\ &= I [TS(T)]_{T_h}^{T_c} - I \int_{T_h}^{T_c} S(T) dT \\ &= I(T_c S(T_c) - T_h S(T_h)) + I \int_{T_c}^{T_h} S(T) dT, \end{aligned} \quad [\text{S36}]$$

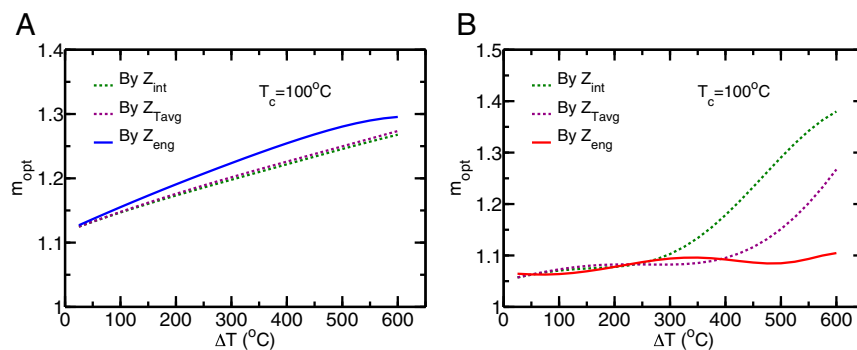
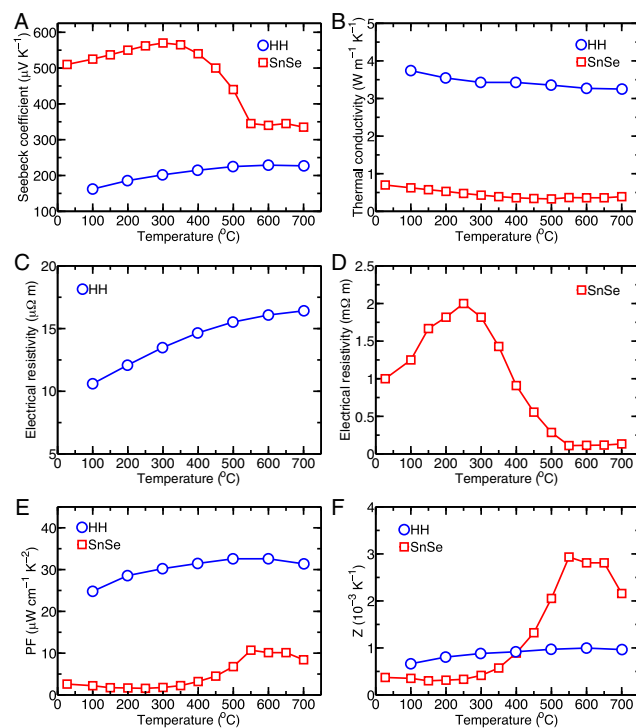
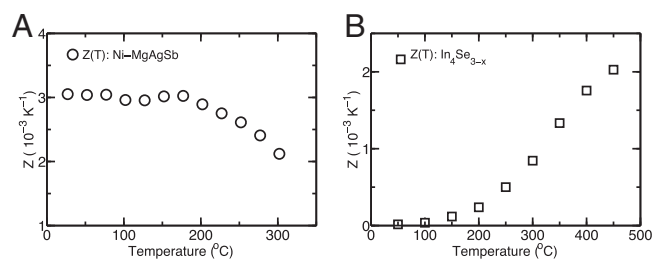
where electric current I is

$$I = \frac{V_{oc}}{R(1+m)} = \frac{\int_{T_c}^{T_h} S(T) dT}{R(1+m)}. \quad [\text{S37}]$$

Substituting Eqs. S36 and S37 into Eq. S35 yields

$$\begin{aligned} P_{\text{out}} &= I \int_{T_c}^{T_h} S(T) dT - I^2 R \\ &= \frac{\left(\int_{T_c}^{T_h} S(T) dT \right)^2}{R(1+m)} - \frac{\left(\int_{T_c}^{T_h} S(T) dT \right)^2}{R(1+m)^2} \\ &= \frac{\left(\int_{T_c}^{T_h} S(T) dT \right)^2}{R} \frac{m}{(1+m)^2} \\ &= \frac{V_{oc}^2}{R} \frac{m}{(1+m)^2}. \end{aligned} \quad [\text{S38}]$$

Therefore, Eq. S38 based on integrating the Seebeck coefficient for the open-circuit voltage is the analytical expression accounting for the Thomson effect for the output power generation, as shown in Eqs. S1 and S23.



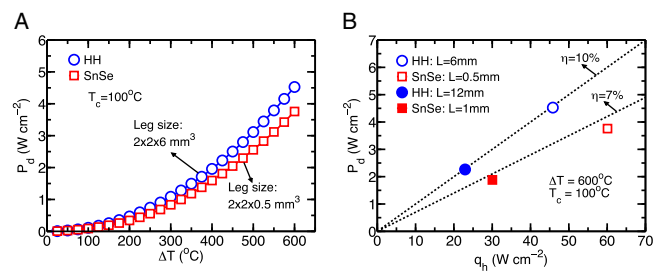


Fig. S4. (A) Comparable output power density by numerical simulation with different leg length of HH and SnSe. (B) Input heat flux vs. output power density in various leg lengths of HH and SnSe where $\Delta T = 600^\circ\text{C}$ and $T_c = 100^\circ\text{C}$. The dashed lines represent conversion efficiency of 10% and 7%, respectively.